

Designing Navigation Tool for Immersive Analytics in AR

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ABSTRACT

As augmented and virtual reality technologies continue to mature, many studies have looked into navigation in 3D visualizations in recent years. However, most studies that examine navigation in 3D visualizations were conducted using a virtual reality environment (VR). Navigation in augmented reality (AR) may have different needs and challenges which have remained undiscovered. My research aims to investigate the elements that influence navigation performance in AR and how to improve the efficiency and user experience of navigation in AR by design, development, and user studies of a navigation tool EaseNav that supports users' navigation procedure.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction techniques—Gestural input; Human-centered computing—Interaction design—Empirical studies in interaction design

1 INTRODUCTION

With many recent 3D navigation studies conducted in VR environments [1, 8, 10], it is time to explore what users need for AR navigation and whether VR navigation techniques transfer well when used in AR. With the visibility of the real world, we anticipate that the color/context of the user's space will influence participants' perceptions of the environment [4, 9], which could increase the difficulty of navigation in AR. However, the visibility of the real world could increase users' situational awareness, which can facilitate navigation with the surrounding knowledge from the real world [9]. On the other hand, cluttered backgrounds in AR might affect users' performance [6]. Many research questions must be answered before designing navigation techniques and improving navigation efficiency in AR environments.

Navigation includes two components: travel and wayfinding [2]. Wayfinding is a cognitive process of making decisions about the route to the destination. Travel is the motor component that allows the user to control the position and orientation of the viewpoint. Wayfinding requires the user to have spatial knowledge, including knowledge of their surrounding environment. Travel can be achieved through simple physical motions such as walking or other travel techniques such as teleportation. Compared to physical movement during exploration, teleportation assists users in translating to a specified destination instantly. However, it can cause spatial disorientation and discontinuity, which negatively influences the user's mental model. Previous work explored using "minimap" for teleportation in VR so that users were able to be teleported to a location outside of their field of view (FOV) [10]. Yet the teleportation in "minimap" could lead to greater context-switching costs due to a lack of spatial context during teleportation and decreased knowledge of the surrounding environment. Meanwhile, a prior study showed that users have more physical motion in AR, possibly because of increased situational awareness [9].

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Researchers found that a large fully-immersed environment has drawbacks such as lacking overview and blind spots [5]. The survey results of IA showed that context-aware navigation techniques such as World In Miniature (WIM) were not exploited enough, even though context awareness is critical for the user while the user is exploring large-scale data visualizations in an immersive environment [3]. Also, based on prior works [9, 11], a single view can lead to inaccurate depth estimates due to the insensitivity of the human eye to the perception of depth along the direction of the version. For mitigating the error, studies showed that providing multiple views can benefit the correct perceptive projection development [11]. Or users would spend more time and energy examining the visualization from different viewpoints on navigating around the visualization to ensure the accuracy of position estimation [9]. Therefore, offering users WIM with multiple views of the 3D visualization can help users achieve efficient and accurate navigation in the AR environment.

2 RESEARCH DESIGN

My research aims to investigate how researchers can utilize what we already know about AR environments and the navigation techniques in VR to design more suitable navigation approaches and tools for supporting users' navigation in AR environments. Below I depict the blueprint of my studies towards the goal: developing a navigation tool that can assist users during the navigation process with minimum distraction, a user study to evaluate the navigation tool with different navigation approaches, and iterative design of the tool through multiple small user studies and improve the usability of the tool according to users' feedback and quantitative measurements.

2.1 Navigation Tool: EaseNav

Based on our preliminary study results from an interaction comparison study with immersive visualization in AR and VR environments (Figure 1), users have more difficulty performing immersive analysis tasks in AR than in VR. It might be due to the cluttered environment and/or the interfered color perception because of the visibility of the real world. Moreover, we noticed that users rotated visualization often to access data points from different sides of the visualization due to occlusion and depth estimation. Besides the uneasy-to-use interaction techniques with the AR head-mounted devices, we think another reason that caused users' frustration was the constant need for navigation through the visualization with immature interaction techniques.

To improve the efficiency and user experience for navigation in AR, I will create a navigation tool (EaseNav) based on the concept of WIM and easy-to-perform interaction techniques for using WIM to navigate. I will design interaction techniques to switch between views using minimum effort so that users will get the least distraction from wayfinding. To eliminate the occlusion, the default presentation in the navigation procedure will have no WIM view. When users perform an open hand gesture, a view from the top of the visualization will show up, which can present the target location on the X and Y axes. To acquire views from other sides, users can perform pinch gestures with thumb and index finger or thumb and middle finger to access the view of axes Y, Z, and X, Z, respectively. Moreover, users could use multiple views to double-check if they find the correct targets or a mismatch because of the single line of sight. To prevent the color in the physical world from interfering

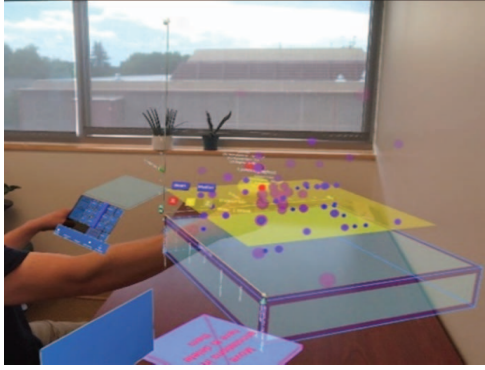


Figure 1: The Immersive Visualization Used in the Interaction Comparison Study

with the view of the WIM, the presented view will have solid color as the background.

2.2 Navigation in AR

I will evaluate EaseNav with four conditions. They are three different navigation approaches assisted by EaseNav: visualization manipulation, locomotion through a controller, and locomotion through teleportation. As the baseline, the fourth condition is locomotion through a controller without EaseNav. Aside from assessing the performance while using EaseNav and getting users' feedback, this experiment also investigates different navigation techniques and the navigation tasks they are suitable for. As an example, when users prefer visualization manipulation over locomotion through a controller? How can I improve the user experience when they locomote through a controller? With less concern about motion sickness in AR locomotion, using a joystick may be possible [7]. Teleportation is an option for alternative efficient navigation approaches. We anticipate that the spatial disorientation and context-switching costs with teleportation in VR will be mitigated in AR due to the visibility of physical space in the real world. I aim to answer the following research questions: RQ1) How does EaseNav support the navigation process in AR? RQ2) Which navigation approach is suitable for performing navigation tasks in AR?

2.3 Iterative Design of EaseNav

Based on the first user study result, I will choose the most suitable navigation approach for the AR environment or implement different navigation techniques based on certain tasks and scenarios to improve the usability of EaseNav. For example, if the input data are small, the system will reduce the default size of EaseNav accordingly, and vice versa. We also can provide options for users to customize their EaseNav based on individual preferences such as handedness, and real-world conditions, such as brightness of the environment or clutter of the surrounding, so that users can adjust the background color or size of WIM. Moreover, I imagine that with a more complex visualization, navigation efficiency could be heavily influenced by the wayfinding process and also individual spatial ability. Therefore, providing users with navigation aids and cues in WIM could be very helpful. For example, adding cues such as spotlights or arrows along the recommended path could prevent users from losing track of the path, especially when the travel distance is long. Furthermore, with more realistic scenarios, revisiting could be another common need for navigation in visualization. Different navigation techniques and aid interactions design could affect the recalling ability of users. As a result, I will adopt the iterative design method to conduct small user studies and continually improve the design of EaseNav. Through those studies, I aim to answer the

following research questions: RQ4) What customization features do users need? RQ3) What kind of cues in AR can help users during navigation with the least distraction? RQ4) What kind of navigation techniques can enhance users' capacity to recall?

3 CONCLUSION

Through my studies, we will have a deeper understanding of navigation in AR, different navigation techniques, performance and usage in AR, and most importantly, a highly functional navigation tool to assist users in accessing targets efficiently and accurately. I hypothesize that EaseNav will increase the popularity of AR devices among consumers since navigation is a common task in both professional and daily life. Below are the questions I would like to discuss and get feedback: 1) What kind of tasks should I include in my user studies for the tool EaseNav? 2) Are there other aspects I should consider when designing the EaseNav? 3) What are other potential needs for navigation in future AR applications?

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